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RESEARCH ARTICLE

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Balanced cable routing for offshore wind farms with obstacles

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Abstract

The *cable routing problem* is the task of designing the minimum-cost electrical network between the turbines of offshore wind farms while satisfying several constraints. It contributes to decreasing wind energy costs, favoring the green transition. This paper focuses on balanced cable routing, studying both branched and radial network topology. First, we propose a visibility graph to route cables around obstacles and boundaries present in the wind farm area. Next, we develop a novel Large Neighborhood Search (LNS) heuristic using neighborhoods of increasing complexity: *swap*, exchanging two turbines between root-branches; *double swap*, exchanging two pairs of turbines; *cycle swap*, moving turbines between root-branches; and *re-partition*, re-partitioning and re-routing two adjacent root-branches. We also extend the LNS heuristic to the unbalanced branched configuration to compare to existing literature. On average, it finds solutions of the same quality using two orders of magnitude less run-time, improving the solutions in five cases.

KEYWORDS

large neighborhood search, math-heuristic, network design, offshore wind farms, sweep algorithm, visibility graph

1 | INTRODUCTION

An offshore wind farm is typically designed in three main steps: first a proper area is selected, second the turbines are located in the area, and finally turbines are connected with cables. The *cable routing problem* consists in finding the optimal network of inter-array cables, that transfers the electricity produced by the wind turbines. The electricity is conveyed to a central hub, the *substation*, which is connected onshore by a high-capacity export cable. The cable routing problem is known to be NP-hard even with only one substation [10]. Large wind farms often need more than one substation, thus making the problem even more challenging.

Because wind parks are increasing in area extension and more complex sites are now being considered, *obstacles* are much more frequent than in older wind farms. Obstacles may occur due to, for example, seabed conditions, existing pipelines, infrastructures, ship wrecks, and regulations, such as navigational and fishing corridors. Cables are not allowed to be placed in these areas, making the optimization more challenging. The inter-array cables that connect the turbines are expensive both to manufacture and install [12], representing more than 5% of the total capital expenditure (CAPEX) of an offshore wind farm [3]. The industry trend in upcoming wind farms is to locate more turbines in a wider area, which will lead to an increase in cable lengths and in their associated costs. For these reasons, the optimization of the inter-array cable configuration is important to lower the levelized cost of energy (LCOE) of offshore wind farms and to enable wind energy to reach zero-subsidy bids [8]. Once this point is reached, a wind farm will become a self-supporting investment, which will promote the green transition and help Europe to reach its renewable energy targets of increasing the offshore wind capacity of at least 76 GW by 2030 [22]. In

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the cable routing problem, the objective is to minimize the total cable costs. A given set of *cable types* is available, each type having a cost (per meter) and a capacity. In addition, *cables crossings*, while technically feasible, are forbidden since they result in increased installation costs and higher risk of failures [2].

1.1 | Balanced cable routing

The inter-array cable routing problem has been explored in literature for different configurations. As illustrated in Figure 1, there are three main schemes for connecting the turbines depending on the inter-array cable topology: branched, radial, and looped.

The first topology, *branched*, allows multiple cables to enter a turbine and exactly one cable to exit the turbine. The *radial* topology is a subset of the branched topology since it connects turbines into “strings,” in which at most one cable can enter and exactly one cable must exit the turbine. Since the radial topology is a subset of the branched one, the resulting cable routing cannot be shorter than using a branched topology. Finally, a *looped* connection is similar to the radial one but with closed strings. This introduces redundancy in the system, making it more robust in case of failures. For an overview of the different designs for inter-array cables, we refer the reader to Pérez-Rúa and Cutululis [19] as well as the detailed study in Fischetti and Pisinger [9]. We refer the reader to Lundberg [18] for an in-depth analysis of the electrical configurations of a wind farm and to Hou et al. [13] for a review on offshore wind farm optimization.

For a branched or rooted topology, each cable leaving the substation has an associated network that we denote a *root-branch*. A cable routing is *balanced* if all root-branches contain the same number of turbines, or differ at most by one in case the number of root-branches is not a divisor of the number of turbines.

In this article, we focus on *balanced* configurations, in which the number of turbines in each root-branch is equalized. This configuration is widely used in the offshore wind farm industry [1]. Receiving the same electrical loads from each root-branch allows the transformers in the substation to share the same design, which presents the following advantages:

- The design and parameter testing phases of the transformers can be standardized, thus reducing installation costs and labor time.
- A single type of transformer is needed as spare in case of failures. If the spare transformer is already installed in the substation (*online spare*), it can substitute for any of the other transformers in case of failure, since they share the same specifications. Due to spatial limitations, it is frequently impossible to accommodate more than one spare transformer in the substation.
- It balances the wear of the components, thus it is preferred for the lifetime of the transformers.
- The protection schemes are easier to design for a set of identical transformers.

For these reasons, the balanced configuration is preferred in practical applications by offshore wind farm developers, mainly because it reduces costs of spare components and of maintenance in case of failures. In fact, removing the balanced constraint will lead to less expensive cabling solutions, but these savings do not offset the additional costs of different transformers and spare parts. Indeed, the unit and installation cost of a single transformer is around 5M€, depending on its rating [23].

1.2 | Related works

The *balanced* cable routing problem has not been studied in literature as much as the *unbalanced*. The *radial* configuration problem, in which turbines are arranged into strings, has been studied only in the *unbalanced* case. Bauer and Lysgaard [2] proposed a hop-indexed integer programming formulation, with planarity constraints to forbid cable crossings. These planarity

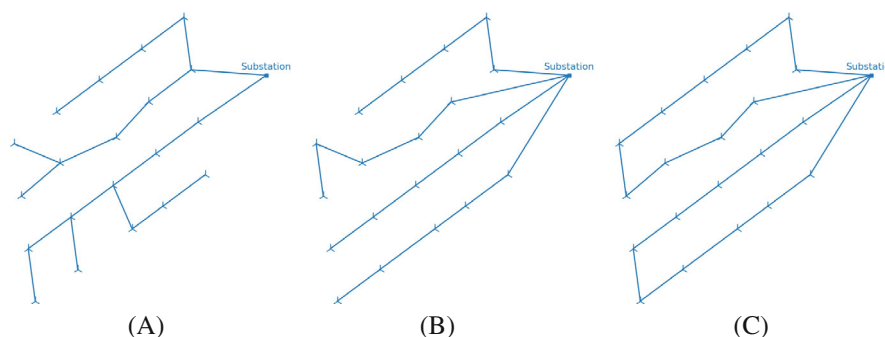


FIGURE 1 Examples showing the three studied cable routing configurations for a wind farm. (A) Branched. (B) Radial. (C) Looped

constraints are excessive even in small cases, so they adapt the Clark and Wright saving heuristic [6], used for the vehicle routing problem, to what they call a planar open savings heuristic. They show that this heuristic is on average only 2% more expensive than the optimal layouts. Katsouris extends the planar open savings heuristics and compares in more detail its variants [17]. Jenkins et al. [14] develop a greedy algorithm for the *unbalanced* configurations to find an initial solution, which is then optimized using a genetic algorithm. They apply it to both *radial* and *branched* designs, and compare it to hand-constructed solutions. They note that a balanced solution is preferable, but do not include the constraint in their model. Fischetti and Pisinger [9] develop a math-heuristic approach, combining a heuristic with a MILP formulation of the problem, for the *unbalanced branched* topology and extend it to both the *radial* and the *looped* configurations. Gong et al. [11] investigate a *unbalanced branched, radial, and looped* topology including the 3D variations due to seabed conditions. Cazzaro et al. [4] compare various metaheuristic for the *unbalanced branched* configuration, concluding that Variable Neighborhood Search performs the best for this problem. Yi et al. [24] also optimize the inter-array cables, combining greedy and stochastic methods for a *branched* topology. Finally, Jin et al. [15] include power loss costs in the optimization for the *unbalanced branched* configuration.

None of the mentioned papers enforce the balanced constraint, which is relevant in practical applications and is the focus of our study.

Even though avoidance of *obstacles* is crucial in practical applications, few papers take them into consideration. Obstacles are becoming more common in new wind farms, as they cover larger areas and are becoming more constrained by existing infrastructures at sea. Yi et al. [24] compute the visibility graph in order to bypass these exclusion zones. They also highlight the need to treat special cases of cable crossings due to the use of the visibility graph. Fischetti and Pisinger [10] use a different approach, by introducing additional points, called *Steiner points*, that are added near existing obstacles in order to make it possible for cables to circumvent the obstacles. They extend the MILP formulation to utilize these optional points. Pillai et al. [20] use a navigational mesh for pathfinding around obstacles, by computing the Delaunay triangulation, for the branched topology. The pathfinding includes Steiner points (which they name Fermat points) to route cables around obstacles. Finally Qi et al. [21] use a Voronoi tessellation to compute the distances between turbines, which can be modified to handle obstacles.

1.3 | Optimization

The objective of the studied optimization is then to find a balanced cable routing while minimizing the cost of the cables. Furthermore, cable crossings are forbidden in our model.

We propose a new way to handle obstacles in the wind farm by using a *visibility graph* to precompute the shortest paths around the obstacles. This approach has the advantage that obstacles can be handled transparently, thus allowing all techniques developed in literature to be applied without changes. In addition, the problem complexity does not increase since there is no need to include Steiner points in the optimization.

As noted in literature, exact approaches to this problem only converge to an optimal solution in a reasonable time for a limited number of turbines [13]. Since the development of offshore wind farms is toward having more and more turbines [13], which is also the case in the instances that we use as benchmark, we develop an heuristic approach for this problem.

We develop a novel large neighborhood search (LNS) heuristic to solve the balanced cable routing problem with obstacles and multiple substations. The LNS heuristic combines purely heuristic methods with exact approaches, gaining advantages from both approaches. To ensure a balanced solution, we split the turbines into root-branches with a uniform number of turbines in each root-branch. The LNS heuristic makes use of a number of neighborhoods of increasing complexity. All neighborhoods ensure that the cardinality of the root-branches is preserved. The neighborhoods are:

- *swap*, exchanging two turbines between root-branches;
- *double swap*, exchanging two pairs of turbines;
- *cycle swap*, moving several turbines between several root-branches;
- *repartition*, taking two adjacent root-branches and completely repartition and reroute the two branches.

The two first neighborhoods are fast to evaluate using brute-force enumeration, since the number of turbines in each root-branch is relatively small. The *cycle-swap* neighborhood makes use of an MIP model, maximizing expected savings by moving turbines to other root-branches, while ensuring that each root-branch will have an unchanged amount of turbines. By construction, the MIP formulation is totally unimodular, and hence can be solved efficiently using an LP solver.

In the *repartition* neighborhood, we iteratively consider two adjacent root-branches and completely repartition and reroute the two branches using an MIP model. As noted by Fischetti and Pisinger [10], the non-crossing constraints are particularly heavy in an MIP formulation of the cable routing problem. Since cable-crossings occur very seldom when considering two

adjacent root-branches, we can avoid the non-crossing constraints, making the model much faster to solve. If our MIP model returns a solution with cable-crossings, we simply reject this candidate solution.

We demonstrate the effectiveness of our LNS heuristic for the balanced problem, studying both radial and branched configurations. We use a set of cable routing cases with a large amount of turbines, multiple substations, and obstacles, in a set of instances that is close to real-life cases in industry. No neighborhood of the LNS heuristic was superior to the other, but, depending on the specific instance, different neighborhoods contributed to improve the solution. To illustrate this, we report the contribution breakdown of each neighborhood both with respect to running time and solution quality for each tested instance. We also extend our LNS heuristic to handle unbalanced problems, in order to make it possible to benchmark against results from the literature. It shows that our LNS heuristic on average is able to find the same quality of solutions as specialized algorithms for the unbalanced problem, and we are even able to improve the best-known solution for 5 instances. Our running times are two orders of magnitude faster than previous approaches. In addition, we report a comparison between the balanced and the unbalanced configuration, quantifying the routing cost difference in the two cases and showing why the balanced configuration is currently preferred in practical applications.

1.4 | Contribution

We summarize the main contributions of this article in the following points:

- We discuss the *balanced* cable routing problem, and explain why it is a common setup in offshore wind farms.
- We propose a preprocessing method to account for obstacles in the wind farm based on a visibility graph. This preprocessing can be applied to other optimization techniques as well with minimal overhead.
- We develop a matheuristic algorithm to solve the problem, which combines heuristic and exact moves.
- We present a computational analysis that breaks down the contribution of each neighborhood to the final solution, for both branched and radial topology.
- We compare balanced and unbalanced configurations of the cable routing problem, quantifying the cost difference between the two.
- We benchmark our optimization technique to state-of-the-art results on the unbalanced branched cable routing.

1.5 | Outline

This article is organized as follows: Section 2 formalizes the balanced cable routing problem. Section 3 describes the methods developed to solve the problem: a preprocessing algorithm for handling obstacles in the wind farm, the initial solution algorithm, and the LNS heuristic. In Section 4, we first benchmark the LNS heuristic for balanced radial and branched topologies, using a set of realistic instances for large parks with obstacles and multiple substations. Since no results on balanced configurations have been reported in the literature, we furthermore benchmark our LNS heuristic on unbalanced branched instances against results from the literature. We also compare the cable routing costs in the balanced and unbalanced cases, in order to estimate their advantages. Finally, Section 5 summarizes the optimization techniques and their results, and provides directions for future studies on the problem.

2 | PROBLEM STATEMENT

In this section, we formally define the balanced cable routing problem of inter-array cables in offshore wind farms.

Let V_s be the set of turbines to connect to substation s , E_s the complete set of edges in $V_s \cup \{s\}$ and K_s the set of cables available to connect the turbines to substation s . For each cable type, a capacity and a cost per meter is defined. By following the approach in [10], the cost of the cables can be preprocessed to include an estimate of the electrical losses in cable. This is possible because we know the nominal production of turbines and how many turbines are connected to each edge in the solution. In this way, the cable losses depend only on the length of the cable, which is already included in our objective function.

The goal of our optimization is to minimize the cost of the cables:

$$\min \sum_{\substack{ij \in E_s \\ k \in K}} d_{ij} c^k x_{ij}^k,$$

where x_{ij}^k is a binary variable that is 1 if we place an edge (cable connection) between points i and j , and k indicates the selected capacity of the cable. Here d_{ij} is the distance of the shortest path between the two points i and j , and c^k is the cost per meter of the cable that can support k turbines. The constraints are:

1. Topology:

- (a) *Branched*: A turbine can have multiple entering cables, and exactly one exit cable, thus turbines are connected in a tree.
 - (b) *Radial*: All turbines have at most one entering cable, and exactly one exit cable, thus turbines are connected in strings.
2. *Balanced*: The number of turbines in each root-branch can differ by at most one turbine.
 3. *Flow conservation*: All the electricity produced by the turbines must be transferred to the substations. In addition, the cable type used in each connection must have enough capacity to transfer the electricity produced by its turbine plus all upstream turbines in its string.
 4. *No crossings*: Cables cannot cross each other.
 5. *Obstacles/boundaries*: Cables cannot be placed inside obstacles nor outside boundaries.
 6. *Substation connections*: The substation must be connected with C root-branches.

3 | SOLUTION APPROACH

This section describes in detail our LNS heuristic to solve the balanced cable routing problem. First, we propose a novel approach to consider obstacles and boundaries in the optimization, without making the problem harder to solve. Second, we discuss a model to partition the turbines when multiple substations are present in the wind farm. Third, the LNS heuristic is discussed, divided into its components: the initial solution method and the various neighborhoods. We highlight how the heuristic works for both branched and radial designs with minor modifications. We detail its heuristic and exact neighborhoods and we discuss how those contribute to the overall heuristic. Finally, we extend the LNS to unbalanced solutions, in order to be able to benchmark against results from the literature.

3.1 | Obstacles and boundaries

Obstacles are present in nearly all new wind farms, increasing the complexity of finding the optimal configurations for the inter-array cables. We define an obstacle as a closed polygon, which is defined by a series of 2D points that are connected together in clockwise order. Obstacles can be both concave and convex. Cables are not allowed to pass inside obstacles; if they do the solution is not feasible. In addition, practical applications also define outer *boundaries*: cables are not allowed to exit from the boundary polygons.

We assume that obstacles and boundaries are part of the input, together with the positions of the turbines and of the substations. One option is to forbid all connections that cross an obstacle. In this way, though, the solution space is reduced and we could potentially forbid cheaper cable routing solutions. For example, if a large obstacle lies between the substation and all turbines, this approach will forbid all connections to the substation, and thus no solution would exist. For this reason, instead of forbidding the connection, we compute the shortest path around the obstacle, thus allowing a feasible solution in this case and in general without limiting the solution space. To find the shortest connections of the cables around obstacles, we compute the *visibility graph*, similar to the approach in [20,24]. A visibility graph connects V vertices that are “visible” one another. If the straight-line connection between two vertices crosses an obstacle or a boundary, the two vertices are not visible. In our definition of the visibility graph, we allow for collinearity, meaning that a connection between two vertices can be exactly on top of an edge of an obstacle.

Having computed the visibility graph between all the turbines, the substation(s), and the points defining obstacles and boundaries, we find the shortest path from each turbine to all other turbines and substation(s) using Dijkstra’s algorithm. Since Dijkstra computes the shortest path from one node to all other nodes, we only need to call the algorithm $|V|$ times. A subset of the shortest paths is shown in Figure 2, using the substation as root.

In order to make obstacles and boundaries transparent to the optimization, we store the shortest path distances in an augmented distance matrix. An auxiliary matrix keeps track of arcs used in the shortest path for each pair of points. These two data structures capture the information of the shortest paths, thus the visibility graph and the shortest paths are computed only once, in a preprocessing step. In this way, most techniques for optimizing the cable routing can be applied to a layout with obstacles with minimal overhead.

One drawback of the visibility graph is in its implementation, because testing visibility is complex and special care is needed to handle edge cases of cables crossings. As Yi et al. [24] note, cables that run along one obstacle edge may lead to a crossing in a special configuration, which must be handled separately with an additional checks.

Finally, in practical applications, the cables need to respect a safety distance from turbines and from other cables. To handle this requirement, a refinement phase is performed at the end of the optimization.

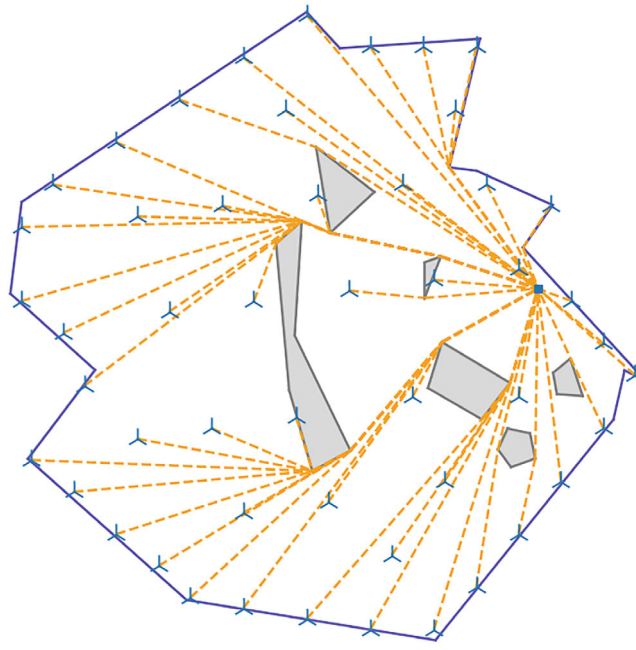


FIGURE 2 An example of the shortest paths computed using the visibility graph in a wind farm. The substation (indicated by the blue square) is used as root. The purple outline is the boundary of the wind farm, while the gray areas are the obstacles which cannot be crossed by cables. The dashed lines are shortest paths from all turbines to the substation

3.2 | Turbine partitioning

An additional problem arises with multiple substations: for each turbine we need to decide which substation it should be connected to. We solve this problem as a first step, before applying the heuristic discussed below. To solve the turbine partitioning, a binary variable is introduced:

$$x_{js} = \begin{cases} 1 & \text{if turbine } j \text{ is assigned to substation } s, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

We are given, for each substation s in the set of all substations S , its total number of turbines to support k_s . Furthermore, J is the set of turbines to connect. The partitioning problem is formulated as follows:

$$\min \sum_s \sum_j d_{js} x_{js}, \quad (2)$$

$$\text{s.t. } \sum_s x_{js} = 1 \quad \forall j \in J, \quad (3)$$

$$\sum_j x_{js} = k_s \quad \forall s \in S, \quad (4)$$

$$x_{js} \in \{0, 1\} \quad \forall j \in J, s \in S, \quad (5)$$

where the objective is to minimize the total sum of the shortest path lengths of each turbine to a substation. Thanks to the augmented distance matrix, we take into account the obstacles in the area without needing additional constraints or special cases. The first constraint in the formulation restricts each turbine to be assigned to one substation only, and the second constraint guarantees that each substation will support the desired total number of turbines.

After the turbine partitioning is calculated, the LNS heuristic is applied to each substation independently.

3.3 | LNS heuristic

Our solution method is an LNS heuristic, where we alternate between different intensification and differentiation neighborhoods. The heuristic combines purely heuristic neighborhoods and exact solution methods.

Figure 3 gives an overview of the different components of the LNS heuristic. First, we have the preprocessing step to handle obstacles and boundaries computes the shortest paths for all turbines. Second, for multiple substations, the partitioning problem

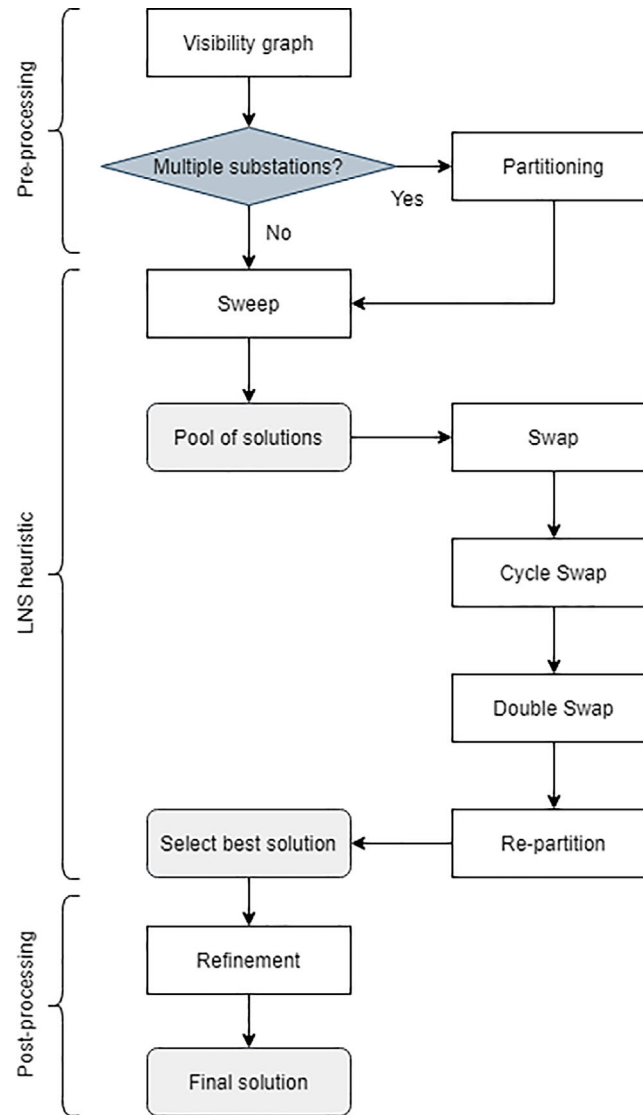


FIGURE 3 Overview of the overall algorithm to optimize the balanced cable routing problem, which shows the various components of the developed large neighborhood search heuristic

is solved to assign turbines to each substation. The LNS heuristic starts by generating a pool of initial solutions using the modified *sweep* algorithm. Then the *swap* neighborhood search is used to improve the solutions in the pool. Next, the *cycle swap* exact neighborhood search is applied on the solutions, followed by the *double swap* which is computationally more expensive. Finally, the *repartition* exact neighborhood search, based on an MIP model, is applied on the solution pool until the solutions cannot be improved anymore. Our LNS heuristic stops at the end of this process and returns the best solution in the pool. A final post-processing step is applied to the best solution, to move slightly the cables in order to respect safety distances that are needed for practical applications.

As we highlight in the next sections, we can find quickly an inter-connection between the turbines of each root-branch. The cable routing problem then in practice reduces to find the optimal division of turbines into root-branches. This key observation is exploited when building the pool of initial solutions and in the various components of our heuristic. Indeed, both the local search and the differentiation phases of our heuristic modify the root-branches, which is different than, for example, using moving single the cables to improve the solution [4] because it is more effective in moving from a feasible solution to another and thus escaping local minima during the optimization while maintaining feasibility.

3.4 | Initial solution

Generating an initial solution for the cable routing problem is a challenging task, because we need to respect both that the solution is balanced and that no crossings are introduced. Given the total number of turbines to connect to a substation, we compute how many turbines to place in each root-branch, such that the distribution is as balanced as possible, and the maximum capacity of

cables is respected. Formally, let S_i indicate the number of root-branches containing i turbines, and p be the maximum number of turbines supported by the cable set. Then we set $S_p := \lfloor n_t / |C| \rfloor$ and $S_{p-1} := n_t \bmod p$, where n_t is the total number of turbines and $|C|$ is the total number of root-branches connected to the substation.

We generate an initial solution by modifying the *sweep* algorithm proposed in [4]. The core idea is to order the turbines by angle with the substation. The ordering is used to partition the turbines into sets (root-branches), thus exploiting the planar nature of the problem. The turbines in each root-branch are connected to the substation using either a string or a branch, and the solution is evaluated.

The original algorithm is modified to consider obstacles and boundaries in the wind farm. In particular, the angles used in sweep to find an ordering of the turbines are computed recursively when moving down through the tree. More specifically, we follow the shortest path from the base station to a turbine, and calculate the angle of every edge on the path, resulting in a list of angles for each turbine. The turbines are next sorted according to their list of angles. In this way turbines are divided into root-branches that are adjacent to each other, thus exploiting the turbine proximity in the cable routing with obstacles. Figure 4 shows one possible division of the turbines into root-branches using the modified sweep algorithm.

In the LNS heuristic, sweep is implemented as a multi-start algorithm to generate a diversified pool of initial solutions. In particular, the balancing requirement ensures that each root-branch have the same number of turbines (plus-minus one). Therefore, we are free to choose the alternation of sets with p and $p - 1$ turbines. For example, the optimal solution may consist of consecutive root-branches with p turbines, followed by a number of consecutive root-branches with $p - 1$ turbines, or any of their permutations. Therefore, sweep tries a significant number of all possible root-branches and in addition chooses different turbines as starting point for the partitioning. In this way, the root-branches are effectively rotated around the substation, thus allowing for a rapid screening of the most promising solutions.

Figure 5 shows the strings placed in an initial solution, based on the root-branches of turbines obtained by the modified sweep algorithm.

Since the ordering of the turbines is based on their physical proximity, sweep generates root-branches that lead to feasible solutions most of the time. Crossings can still appear, so a penalty is added if crossings are present in the solution. Avoiding crossings is indeed one of the major challenges of the cable routing problem, and one of the strengths of this approach.

Because of its effectiveness, sweep is used to generate a pool of initial solutions in our heuristic.

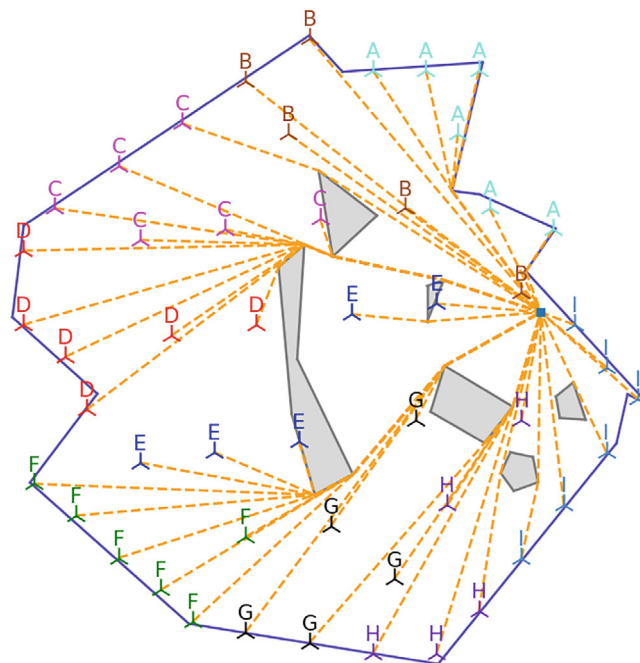


FIGURE 4 A possible balanced division of the turbines into root-branches, obtained by the modified sweep algorithm, which is used to generate one initial solution. Each root-branch has either 5 or 6 turbines, thus satisfying the balancing requirement. The dashed (orange) lines are the shortest paths of each turbine to the substation, which are used to recursively define an ordering of the turbines accounting for obstacles

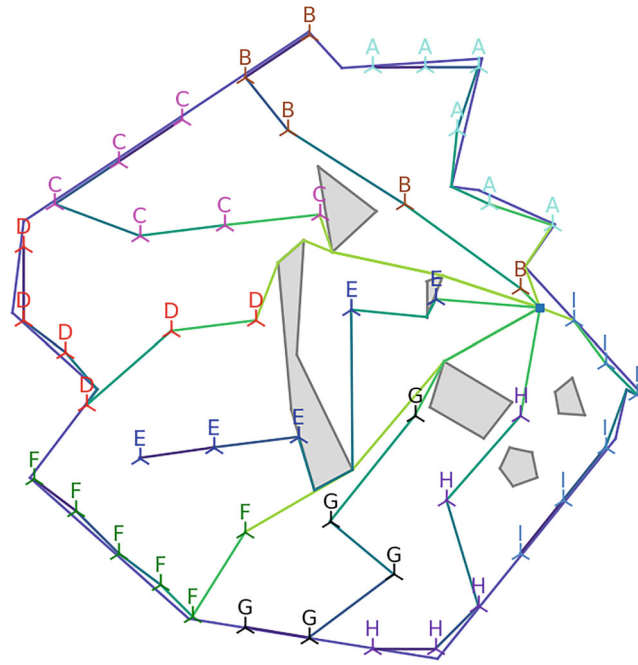


FIGURE 5 A balanced radial configuration with strings placed in each root-branch, as defined by the modified sweep algorithm and denoted in Figure 4. It shows how the cables are routed around the obstacles, thanks to the shortest paths and the augmented distance matrix. Moreover, it is interesting to see that the root-branches definition avoid creating crossings between different strings. The different colors of the cables depend on the number of turbines that each segment supports

3.5 | Inter-connecting a root-branch

Once the partitioning of turbines into root-branches has been found using sweep, the problem is to connect the turbines in each root-branch to the substation, using the branched or the radial topology of minimum cost. Because this mechanism is used extensively during the optimization, both by sweep and in the heuristic, it is a key element for the performance of our heuristic.

Depending on the desired topology, we connect the turbines in a root-branch as follows:

1. *Branched*: We approximate the optimal connection by computing the minimum spanning tree using the Prim–Dijkstra algorithm for evaluating the initial solutions in sweep. The problem is in fact a capacitated minimum spanning tree, which is NP-hard since it contains a special case of the diameter constrained minimum spanning tree (DMST). For our heuristic, we also use a fast heuristic based on cost scaling, which has been used for the capacitated multi-commodity flow problem in Katayama [16].
2. *String*: A multistart greedy heuristic is used, which builds the string starting from the substation. To connect the next turbine, it chooses based on weights, which are based on the physical distances between turbine with some added randomization. The number of tries made by the greedy has been tuned to have a short running time and close to optimal results. In fact, when the number of turbines in the set is low, it is possible to find the optimal string by brute force in a short time.

Thanks to this approach, the optimization can focus on improving the partitioning of the root-branches, which will induce a solution for the cable routing. Indeed, if the optimal partitioning is known, then placing the optimal branch or string in each root-branch will create the routing solution of minimum cost.

3.6 | Local search: Swap

Given an initial solution, a local search considers pairs of adjacent root-branches. We perform a *swap* move, in which two turbines are exchanged between two root-branches. We compute their new cable connections thus created. If the cable routing cost improves, we move to this new solution. In addition, we need to check for crossings: if a crossing is found, the move is rejected. The complexity of a swap move between two root-branches is $O(mn)$, with n being the number of turbines in the first root-branch and m the number of turbines in the second root-branch. If s is the number of root-branches in the cable routing, the total local search procedure is $O(mns)$, since the swap move is symmetric. In practical cases, it is feasible to try all the combinations during the local search.

3.7 | Double swap

An extension of the swap move is to exchange two pairs of turbines at the same time between adjacent root-branches. In this case, the complexity becomes $O(n^2m^2)$ for each pair of root-branches. Therefore, we restrict the candidates for the double swap move to a subset of the turbines in each root-branch. This extension of the swap move is indeed more powerful, especially to diversify the solution, but it is computationally more expensive. A variant, that we also implemented, of the double swap is to exchange two pair of turbines between three root-branches at a time. In this case, the set in the “middle” will exchange two of its turbines, one with the “left” neighboring set and the other with the “right” set. The complexity is $O(lm^2r)$, where l , m , r are the number of turbines for the left, the middle, and the right set, respectively. Both these techniques proved to improve the final solution reached by our heuristic.

3.8 | Cycle swap

Although the swap and double swap heuristics perform well, in some cases it is necessary to move a whole cycle of turbines among root-branches. Therefore we introduce the *cycle swap* move, which is solved using an exact approach. Intuitively, we want to individuate the turbines that will lead to a cheaper solution when moved from their current root-branch. As opposed to the double swap, this exchange is not limited to a sub-set of root-branches, thus allowing to find *cycles*, or rotations, of turbines that improve the cable routing. Starting from the current solution, we compute a move matrix A and a cost vector c . The idea is to capture the potential improvement of moving one turbine alone to a neighboring root-branch, without considering its replacement yet. The cost vector c captures an estimated cost of such move. Say that the best improvement is to move turbine t from set a to set b . The improvement is estimated by computing the difference in length of the original strings (or branches) a and b with respect to the new strings (or branches) a' and b' .

This problem is formulated as an MIP model, named *cycle swap*, to find the optimal assignment of turbines into sets. The binary variable x_j indicates whether turbine j is chosen to move from its current set a to set b . Matrix A then encodes this possible movement, by setting to $A_{ia} = 1$ and $A_{ib} = -1$.

The move matrix A is in general defined as:

$$A_{ij} = \begin{cases} 1 & \text{if turbine } j \text{ will move to set } i, \\ -1 & \text{if turbine } j \text{ will move from set } i, \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

which is totally unimodular by construction. The cost associated with moving this turbine, c_i , is just an estimate of the overall cable routing costs, because by moving one turbine we are not taking into account which other turbine need to be exchanged to keep the root-branches balanced. This allows us to individuate the most promising turbines to cycle swap, thus considering a wider range of possibilities than with the swap and double swap moves. This leads to the following MIP formulation:

$$\min \sum_{j \in V} c_j x_j, \quad (7)$$

$$\text{s.t. } \sum_{j \in V} A_{ij} x_j = 0 \quad \forall i \in V, \quad (8)$$

$$\sum_{j \in V} x_j \geq 1, \quad (9)$$

$$x_j \in \{0, 1\} \quad j \in V, \quad (10)$$

where j is the index for turbines and i is the index for root-branches. The c_j costs are estimating the improvement of exchanging one turbine, as described above. The objective function minimizes the estimated costs of the moves, and the constraint $Ax = 0$ ensures that the number of turbines in each root-branch is unchanged. The first constraint ensures that the solution remains balanced, because for each x_j moved to another set, it will leave a -1 that has to be filled by another move. The second constraint is added to ensure that at least one turbine (actually two since the first constraint guarantees the balancing) is moved with respect to the initial solution.

The model can be efficiently solved with an LP solve, since A is totally unimodular. However, the preprocessing phase, that estimates the best move for each turbine, has complexity $O(st)$, where s is the number of root-branches and t the total number of turbines. In practice, the need to evaluate the optimal string or branch for each possible move makes it more computationally expensive than the swap moves, especially for larger sets.

3.9 | Repartition

The final neighborhood for the balanced cable routing is called *repartition* and uses a math-heuristic approach. As the name implies, math-heuristics combine techniques for meta-heuristics and mathematical programming (MIP) techniques.

The repartition neighborhood uses an MIP model to find the cheapest cabling of two adjacent root-branches, allowing any number of turbines to move between the sets, but maintaining the cardinality of each root-branch.

The local search of this neighborhood iterates through pairs of root-branches in the solution, until no more improvements can be found, meaning that a local minimum has been reached. Since the MIP model is considering only two root-branches at a time, it is small and fast to solve to optimality. The approach for the radial and the branched topologies is the same, with the radial configuration requiring only one additional set of constraints in the MIP formulation, which we detail below.

Different from other approaches, we omit non-crossing constraints in the MIP formulation, since crossings occur very seldom when we only consider two adjacent root-branches.

3.9.1 | Branched topology

We describe the MIP formulation for connecting turbines in two adjacent sets, using the branched topology. The first formulation is for the *unbalanced* case, that we will use for benchmarking against results from the literature. In the model, s is the index of the substation. We use the binary variable x_{ij}^k to indicate whether a cable with capacity k is placed between i and j , and the continuous variable f_{ij} to represent the flow on the cable.

$$\min \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} d_{ij} c^k x_{ij}^k, \quad (11)$$

$$\text{s.t. } \sum_{k \in K} x_{ij}^k \leq 1 \quad \forall i, j \in N, \quad (12)$$

$$\sum_{j \in N} \sum_{k \in K} x_{ij}^k = 1 \quad \forall i \in N, i \neq s, \quad (13)$$

$$\sum_{j \in N} \sum_{k \in K} x_{sj}^k = 0, \quad (14)$$

$$x_{ii}^k = 0 \quad \forall i \in N, \quad (15)$$

$$f_{ij} \leq \sum_{k \in K} k x_{ij}^k \quad \forall i, j \in N, \quad (16)$$

$$\sum_{j \in N} f_{ij} = \sum_{l \in N} f_{li} + 1 \quad \forall i \in N, i \neq s, \quad (17)$$

$$\sum_{i \in N} f_{is} = N - 1, \quad (18)$$

$$\sum_{i \in N} \sum_{k \in K} x_{is}^k = 2, \quad (19)$$

$$x_{ij}^k \in \{0, 1\} \quad \forall i, j \in N, k \in K, \quad (20)$$

$$f_{ij} \geq 0 \quad \forall i, j \in N. \quad (21)$$

The objective function (11) minimizes the cost of the cable types chosen in the solution, based on the shortest paths d_{ij} between the two turbines and the cost per meter c^k of the cable with capacity k . The constraint (12) ensures that only one cable type is placed between two turbines, (13) allows only one cable to exit from each turbine and (14) that no cable exits the substation. Constraint (15) forbids self-loops of the cables. Constraint (16) connects the flow variables to the selected cable types, making sure that the flow in the cable does not exceed its capacity and (17) ensures the conservation of the flow in the network. Constraint (18) forces all the flow to reach the substation, and (19) constraints the solution to have only two direct connections to the substation. Finally, (20) and (21) define the domain of the variables.

The main advantage of this formulation compared to, for example, [10] is that we have omitted the non-crossing constraints. Since there are KN^4 of these constraints, they significantly increase the size of the model, making it necessary to use branch-and-cut techniques [10] to solve the problem to optimality. In our case, we can omit the non-crossing constraints since

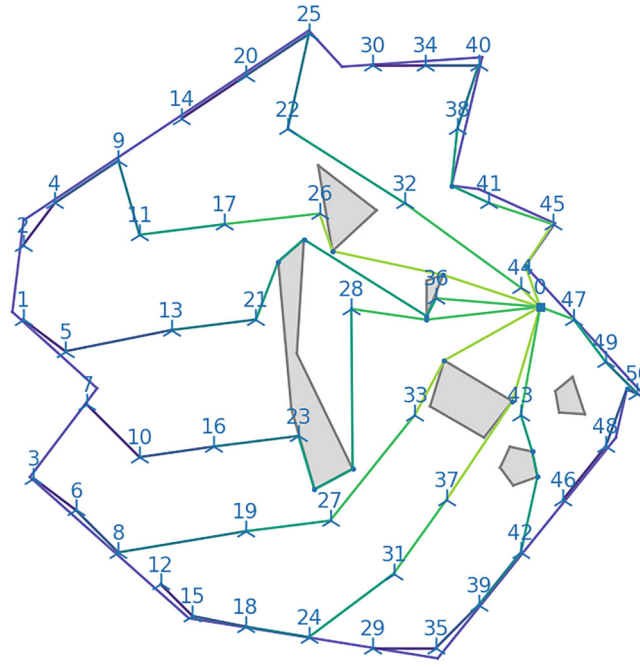


FIGURE 6 An optimized balanced radial solution for a wind farm with 50 turbines (labeled from 1 to 50) and one substation, which is indicated with a square and the 0 label. The single dots are Steiner nodes that are used to bend cables around obstacles, which we handle in the preprocessing phase. It is interesting to see the changes in the routing of the strings compared to the initial solution in Figure 5, especially for turbines that are not directly visible from the substation

we only consider two adjacent root-branches, where crossings very seldom occur. If the MIP returns a solution with crossings, we penalize the found routing significantly to ensure that it is not accepted in the LNS heuristic.

3.9.2 | Balanced constraints

The following constraints are added to the formulation to ensure balanced solutions:

$$0 \leq f_{ij} \leq \left\lceil \frac{N-1}{C} \right\rceil \quad \forall i, j \in N, \quad (22)$$

$$\left\lceil \frac{N-1}{C} \right\rceil \sum_{k \in K} x_{ij}^k \leq f_{ij} \quad \forall i, j \in N, \quad (23)$$

where (22) and (23) ensure that the solution is balanced by, respectively, imposing an upper bound on the allowed flow, and a lower bound on the flow of the chosen cables that enter the substation. C is the number of connections to the substation, which is the number of root-branches in the solution.

3.9.3 | Radial topology

To ensure a radial configuration of the cable routing, an additional constraint is added to the model:

$$\sum_{i \in N} \sum_{k \in K} x_{ij}^k \leq 1 \quad \forall j \in N, j \neq s. \quad (24)$$

The constraint allows at most one cable to enter each turbine, and since constraint (14) furthermore ensures that only one cable can exit a turbine, the turbines are connected in strings.

Figure 6 shows the final routing solution for a wind farm in a balanced radial topology. Compared to the initial solution in Figure 5, it shows the ability of the LNS heuristic to explore the solution space.

4 | COMPUTATIONAL EXPERIMENTS

The computational study is divided into three parts. In the first part, we benchmark our heuristic for the *balanced* problem on a set of realistic instances, showing the contribution of the various neighborhoods to the final solutions. The LNS heuristic is able to find feasible solutions even for wind farms with a high number of turbines, multiple substations and many obstacles. In the second part, we compare the cable routing for the balanced and unbalanced configurations, quantifying the cost difference

between the two cases. Since no results are published in the literature for the balanced configurations, the third part compares our technique to results from the literature on the *unbalanced* branched topology. For the experiments of the second and third parts, we modified the sweep algorithm to generate unbalanced solutions and changed the model formulation of the repartition move.

All the following results were obtained on a Intel® Xeon® Platinum 8168 CPU at 2.70 GHz. The MIP solver used for the repartition neighborhood is Gurobi 9.1.2. LNS terminates when no more improvements can be found, hence no explicit time limit is imposed. The pool of initial solutions is set to 10 for the balanced cable routing and to 15 for the unbalanced problem.

4.1 | Realistic instances

A set of large-scale instances for the wind farm layout problem—choosing the best placement for turbines to maximize their power production—was made available in Cazzaro and Pisinger [5]. Each instance is composed of one or more areas in which the turbines are to be placed. We consider two types of wind turbines, having capacity of 10 and 15 MW, respectively, and optimize the positions of the turbines for each layout. For each area, the high energy density value is used, which results in a very high number of turbines to be connected by the cable routing problem. The generated layouts should thus capture the most challenging cases of cable routing in real wind farms. In each area, multiple substations have been manually added, based on the power production and the number of turbines. An overview of the instances can be seen in Table 1.

Table 2 reports the set of cables available to optimize this set of large instances. There are two cable types for each turbine, and their characteristics (cross-section and cost per meter) originate from [7].

4.2 | Balanced branched topology

Table 3 reports the results of our LNS heuristics on the 15 MW instances for the balanced branched topology. It breakdowns the contribution of the different neighborhoods of our heuristic with respect to the cost of the best solution and of the run time.

Columns refer to the various steps of the LNS heuristic as outlined in Figure 3. Each column reports the cost of the current best cable routing in the solution pool at the end of each neighborhood, and the amount of time spent in the given part of the LNS heuristic. The total run time is reported in the last column, and the final solution cost is the one achieved after the last optimization step, which can be seen in the “repartition” column.

TABLE 1 The considered large-scale instances

Instance	Substations	Turbines per substation	
		15 MW	10 MW
A	2	26, 14	40, 21
B	2	49, 50	75, 74
C	2	60, 30	90, 45
D	3	57, 57, 56	85, 85, 85
E	4	7, 47, 47, 36	10, 71, 70, 55
F	3	26, 66, 66	40, 100, 99
G	3	47, 47, 46	70, 70, 70
H	3	90, 68, 30	119, 118, 45
I	4	82, 77, 77, 77	118, 117, 118, 117
J	5	78, 58, 42, 32, 25	102, 102, 56, 55, 37

Note: The 15 MW instances use a larger turbine, thus a fewer number of them is placed in the same area. For the cable routing, the 10 MW instances are more challenging than the 15 MW instances because of the higher number of turbines to connect.

TABLE 2 The cable set used to benchmark the synthetic instances

Cross-section (mm ²)	€/m	10 MW	15 MW
400	240	6	4
630	336	8	6

Note: Cross section and cost per meter is taken from [7]. Based on the maximum capacity of each cable, we report how many turbines each cable can support, using turbines of 10 and 15 MW capacity.

TABLE 3 Breakdown of the neighborhood contributions by cost and run time on LNS heuristic for balanced branched topology on the 15 MW synthetic instances using the cable set presented in Table 2

Instance	Sweep		Swap		Cycle swap		Double swap		Repartition		Total time (s)
	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	
A	20.60	1	20.39	2	20.39	4	20.39	32	20.31	18	57
B	49.97	4	48.73	4	48.73	12	48.73	80	48.55	55	155
C	47.55	0	46.88	4	46.88	10	46.88	72	46.82	43	130
D	87.03	21	85.10	20	85.10	86	85.10	425	84.40	273	825
E	66.70	2	65.39	9	65.23	28	65.23	173	65.03	79	299
F	85.55	1	83.64	6	83.40	20	83.40	104	83.34	61	193
G	66.91	3	66.36	6	66.36	18	66.36	149	66.19	80	256
H	100.74	3	98.99	9	98.93	28	98.87	164	98.46	85	289
I	178.62	9	175.95	16	175.50	76	175.44	394	175.44	163	656
J	115.27	3	113.28	9	113.28	31	113.28	197	113.16	95	338

TABLE 4 Breakdown of the neighborhood contributions by cost and run time on LNS heuristic for balanced branched topology on the 10 MW synthetic instances using the cable set presented in Table 2

Instance	Sweep		Swap		Cycle swap		Double swap		Repartition		Total time (s)
	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	
A	22.12	1	21.89	3	21.79	8	21.79	56	21.62	36	105
B	55.57	8	55.10	7	54.81	24	54.74	214	54.74	91	344
C	54.39	5	53.63	7	53.60	36	53.59	160	53.49	80	287
D	100.50	12	97.31	17	97.31	50	97.20	423	97.14	151	653
E	76.50	5	74.97	10	74.77	24	74.77	178	74.36	81	302
F	97.26	13	95.94	11	95.50	44	95.50	322	95.43	159	551
G	81.21	1	78.74	11	78.72	28	78.72	216	78.15	98	354
H	120.61	8	117.72	15	117.68	61	117.54	735	117.41	160	981
I	212.53	23	206.24	40	206.20	142	205.93	1231	205.90	283	1718
J	132.88	16	130.25	36	130.25	123	130.02	1871	130.02	466	2514

In general, the solution cost improves throughout all steps of the heuristic. The contribution of the neighborhoods is highly dependent on the characteristics of each instance. In instance H, for example, every neighborhood contributes to improve the cable routing. The biggest improvement are usually achieved in the first phases of the heuristic, especially by swap just after the initial solutions generated by sweep, while the improvements get smaller in the following steps because the solution is more and more optimized. The longest run time is observed for instance D, with 14 min, and the shortest is observed for instance A, with under a minute. The double swap move is clearly the most time consuming phase of the heuristic, accounting for roughly half of the optimization time. The repartition phase is the second most time consuming, since it involves solving several MIP models.

The results for the balanced branched topology on the larger 10 MW instances are shown in Table 4. The number of turbines to connect is about 50% higher than the previous instances, and this is reflected in the run times, which are on average 1.5 times longer than for the 15 MW instances. The worst case, instance J, takes 42 min to connect 352 turbines to 5 substations, which indicates that the heuristic is able to scale satisfactorily even for large wind farms. Also in these cases, all the neighborhoods of the heuristic helps in improving the solution, with instances H and I reaching better solution costs at every step.

4.3 | Balanced radial topology

We break down the contribution of each part of the LNS heuristic for the balanced radial topology. Table 5 reports the results for the 15 MW instances and Table 6 for the 10 MW instances.

For the 15 MW instances, the run time is 3.3 times higher on average in the radial topology compared to the branched case. This is due to the longer time required to connect the turbines of a root-branch into a close-to-optimal string, and indeed the longest running time are for the double swap and the cycle swap neighborhoods, which perform this operation extensively. On the contrary, the additional set of constraints in the MIP model formulation used in the repartition neighborhood does not affect much the computational times, which remain comparable to the branched case.

TABLE 5 Breakdown of the neighborhood contributions by cost and run time on LNS heuristic for balanced radial topology on the 15 MW synthetic instances using the cable set presented in Table 2

Instance	Sweep		Swap		Cycle swap		Double swap		Repartition		Total time (s)
	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	
A	20.76	1	20.56	2	20.56	11	20.56	59	20.56	17	92
B	50.20	17	49.17	32	49.17	144	49.17	434	48.98	53	680
C	48.35	1	47.43	16	47.43	41	47.43	164	47.43	13	235
D	87.47	31	86.07	54	86.07	283	85.98	780	85.98	87	1235
E	70.30	6	67.20	60	67.13	276	67.13	620	67.05	61	1027
F	87.79	2	85.52	43	85.52	95	85.52	317	85.52	31	490
G	67.36	14	66.62	64	66.62	275	66.56	1198	66.56	74	1626
H	101.61	14	99.50	63	99.44	256	99.44	540	99.38	54	928
I	178.98	40	176.22	140	175.98	904	175.98	2078	175.98	163	3325
J	117.27	16	114.81	69	114.81	234	114.81	673	114.81	73	1067

TABLE 6 Breakdown of the neighborhood contributions by cost and run time on LNS heuristic for balanced radial topology on the 10 MW synthetic instances using the cable set presented in Table 2

Instance	Sweep		Swap		Cycle swap		Double swap		Repartition		Total time (s)
	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	Cost (M€)	Time (s)	
A	22.83	21	22.73	56	22.61	528	22.61	1086	22.49	20	1712
B	56.08	189	55.15	206	55.15	2226	55.04	5148	55.00	105	7875
C	55.65	79	55.06	105	54.92	1173	54.92	1897	54.76	46	3300
D	101.40	215	98.54	235	98.54	1582	98.51	3880	98.37	75	5988
E	77.50	77	75.80	331	75.60	976	75.60	1881	75.57	43	3311
F	98.32	209	96.57	215	96.57	1525	96.51	5004	96.51	70	7024
G	81.44	10	78.89	385	78.83	2600	78.82	2593	78.82	63	5652
H	121.69	120	118.54	407	118.47	2480	118.31	7827	118.20	88	10 923
I	213.05	435	207.77	902	207.77	4176	207.77	8867	207.39	148	14 529
J	133.88	243	131.98	462	131.98	2637	131.57	10 351	131.52	105	13 801

The 10 MW turbine instances are more difficult since not only we have 50% more turbines to connect, but the number of root-branches is higher. For these reasons, the running times are significantly longer than for the branched topology and almost 10 times higher than the 15 MW instances. Also in this case, the double swap and the cycle swap moves are the main time consuming components in the heuristic.

In any case, all neighborhoods of the heuristic contribute to improve the final solution, which can be clearly seen in the 10 MW instances C, D, and H.

Compared to the branched topology, the solution costs are higher as expected. The cost gap is of 1.42% on average for the 15 MW instances and of 1.43% for the 10 MW. The relatively low difference in final cost is one of the reasons why the industry prefers the radial topology, which require less electrical components than the branched solutions.

In Figure 7, we show two routing solutions for the balanced radial topology on synthetic instance G, for the 15 and 10 MW turbines. The substations have the same locations in the two cases. All three substations are placed close to a corridor representing the export cable, which cannot be crossed by cables. The total number of turbines to connect is 140 for the larger 15 MW turbines, and 210 for the smaller 10 MW turbines. Even reaching an initial solution can be difficult in these cases, since it is difficult to find a solution without crossings. This instance highlights the advantage of including spatial proximity in the heuristic, both in the modified sweep initial solution, in the root-branch neighborhoods and in the different swap moves.

4.4 | Cost comparison between balanced and unbalanced configurations

The main motivation for a balanced configuration of the cable routing is due to the savings incurred in establishment costs by standardizing transformers in the offshore substation. In this section we quantify the cost difference between balanced and unbalanced topology. As a benchmark, we use the same set of synthetic instances as in the previous section.

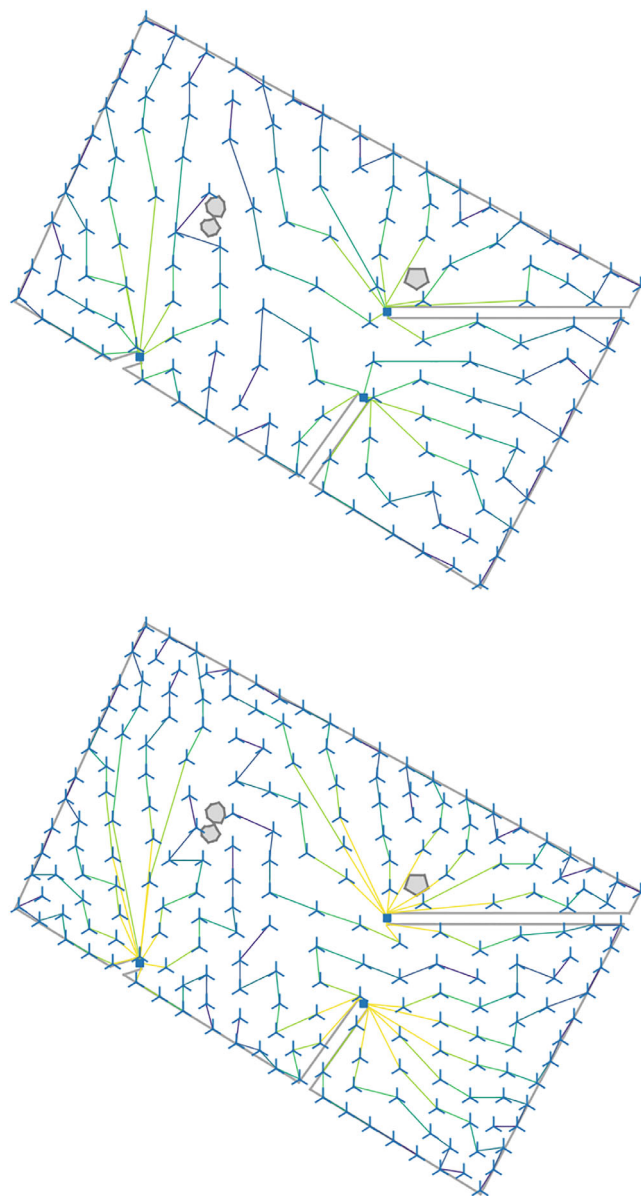


FIGURE 7 Cable routing solution for balanced radial topology on synthetic instance G. On the top figure 140 turbines of 15 MW capacity are connected to 3 substations, for a total of 24 root-branches. On the bottom figure, the same instance G for 10 MW turbines, connecting 210 turbines divided into 30 root-branches

The results comparing balanced and unbalanced configurations for the branched topology are reported in the Appendix. Results for the 15 MW instances are reported in Table A1 and for the 10 MW instances, which are more challenging because of the higher number of turbines, in Table A2. The unbalanced case does indeed reach less expensive solutions than the balanced configuration, saving on average, around 0.5M€ and 2.0M€ for the 15 MW and the 10 MW instances. The results for the radial topology are reported in Tables A3 and A4 for the 15 and 10 MW cases, respectively. In these comparisons, the unbalanced configuration is 0.5M€ less expensive than the balanced case for 15 MW instances, and 1.3M€ cheaper for the 10 MW instances. Surprisingly, in four instances the routing cost is the same, meaning that the balanced solution is possibly optimal even for the unbalanced case.

The highest difference between balanced and unbalanced configurations is 3.4M€ in the tested instances. Since the unit and installation cost of a transformer is about 5M€ [23], the savings of the unbalanced case do not offset the additional electrical equipment at current costs. For this reason, offshore wind farm developers prefer to enforce a balanced cable routing instead of having different transformer ratings, especially in parks with multiple substations.

4.5 | Unbalanced branched topology

To the best of our knowledge, no results have been published in the literature for balanced configurations. Therefore, we modified our LNS heuristic to handle the *unbalanced* branched topology, by modifying the sweep to allow unbalanced solutions and by changing the constraints in the MIP formulation of the repartition neighborhood.

We benchmark our LNS on a set of 29 instances for the unbalanced branched topology, presented in Fischetti and Pisinger [10]. These instances have only one substation each and no obstacles. The original paper uses a math-heuristic approach for the cable routing, solving an MIP model for the whole problem, including no-crossing constraints. Table 7 reports the results of our heuristic compared to the published results in [10].

On average, the solutions found by our LNS heuristic are only 0.01% more expensive than those reported in [10]. In 15 of the 29 instances, the final solution costs are equal, in 12 instances, our solutions are slightly more expensive, and in 5 instances, we improve the best known solutions from the literature.

One clear advantage of the LNS heuristic is the run time: on average it takes about 5 min to converge to the final solution, and 42 min in the worst case. In [10], the time limit was 24 h, and the average run time was almost 700 min.

The LNS heuristic proves to be competitive compared to results from [10]; in the worst case being 0.60% more expensive than the best known solution, and in the best case improving the solution by 1.41%. It is also interesting to see that our heuristic

TABLE 7 Comparing the results on unbalanced branched configuration instances from the literature

Instance	Literature [10]		LNS heuristic		Delta (%)
	Cost (M€)	Time (s)	Cost (M€)	Time (s)	
1	19.44	14 749	19.47	872	0.19
2	21.40	86 401	21.44	1247	0.16
3	22.61	4621	22.62	629	0.05
4	24.45	86 401	24.44	860	−0.02
5	23.48	86 400	23.53	891	0.20
6	24.77	86 401	24.75	1652	−0.09
7	8.56	43	8.56	122	0.00
8	8.81	58	8.81	128	0.00
9	10.06	28	10.06	63	0.00
10	10.30	59	10.30	76	0.00
11	9.20	174	9.20	16	0.00
12	8.60	23	8.60	6	0.00
13	8.93	73	8.93	12	0.00
14	10.17	24	10.17	146	0.00
15	10.35	88	10.35	174	0.00
16	8.05	248	8.05	175	0.00
17	8.56	1278	8.56	198	0.00
18	8.36	354	8.36	141	0.00
19	9.18	1718	9.18	157	0.00
20	38.98	86 400	38.98	178	0.00
21	44.86	86 400	44.83	261	−0.06
22	40.95	86 400	40.95	2491	0.00
23	44.42	86 400	44.54	2314	0.27
24	50.38	86 401	49.67	151	−1.41
25	52.33	86 401	52.43	215	0.19
26	22.34	86 401	22.38	1260	0.19
27	23.36	86 400	23.39	2370	0.13
28	26.64	86 401	26.80	373	0.60
29	27.30	86 401	27.27	523	−0.09
Average		42 522		610	0.01
Median		14 749		198	0.00

Note: The LNS heuristic is benchmarked against the math-heuristic [10]. The lower the cost the better. The bold numbers report the best cost for each instance. Our LNS heuristic finds the same solution cost in 15 out of 29 instances, a more expensive solution in 9 instances and a better solution in 5 instances. On average our technique is only 0.01% more expensive than the results from the literature, using on average 10 min of run time compared to 11.8 h in [10].

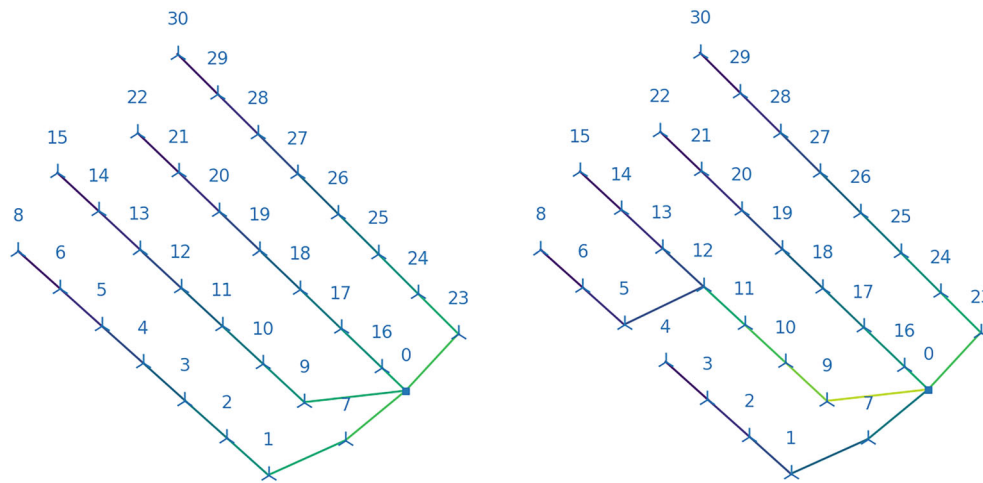


FIGURE 8 Cable routing solution on instance 17 in the literature. On the left, a balanced solution with a total cost of 8.62 M€, and on the right, an unbalanced one with a cost of 8.56 M€

is able to match the optimal solutions reported in [10] on instances 7–19, which have a proven MIP gap less or equal to 0.01% from optimality.

Finally, in Figure 8, we show instance 17 of the literature dataset, in which a balanced branched solution is compared to the unbalanced one. The difference between the balanced and the unbalanced solutions in this case is only of 0.7%. The difference is in the left-most root-branch: in the balanced solution it connects 8 turbines, while the unbalanced solution is using a root-branch of 10 turbines and another of 5, which is an example of a highly unbalanced solution. Interestingly, the balanced solution obtained for the branched topology happens to be also valid for the radial topology.

5 | CONCLUSION

We presented the balanced inter-array cable routing optimization problem, which emerges from practical applications faced by the industry. A novel LNS heuristic was presented using several neighborhoods of increasing complexity, combining heuristic and exact approaches. A preprocessing phase was proposed to deal with obstacles in the wind farm, using a graph formulation based on a visibility graph. The visibility graph determines the shortest paths around obstacles, thus making the obstacle and boundaries transparent for the following optimization part. In addition, the related partitioning problem can make use of the visibility graph to obtain high-quality partitions.

The proposed LNS heuristic makes use of a modified *sweep* algorithm, to generate a set of initial solutions without cable crossings. The LNS heuristic maintains the cardinality of each root-branch by using a number of neighborhoods: the *swap* move, *double swap* move, and the more general *cycle swap*. Finally, the *repartition* move uses an MIP formulation to optimize two adjacent root-branches. These moves are a novel way to approach the cable routing problem compared to existing models in the literature.

The computational results showed that the LNS heuristic is effective in solving very large instances, connecting hundreds of turbines to multiple substations, in a matter of minutes, both for balanced radial and balanced branched topologies. A detailed study of the individual neighborhoods showed that each of the moves contribute to the solution process, and that the neighborhoods are used in increasing order of complexity.

In order to have a comparison with the literature, the heuristic was adapted to handle the unbalanced cable routing configuration. We quantified the cost difference between the balanced and unbalanced cases for both branched and radial topologies on the synthetic instances. The savings of unbalanced solutions, though, do not offset the additional transformer costs needed for an unbalanced routing, motivating why offshore wind farm developers prefer balanced cable routings.

We then compared our heuristic with the literature on unbalanced branched routing. The benchmark showed that the LNS finds solutions of same quality as reported in the literature, while using only a fraction of the time. Furthermore, it is able to improve the best known solution in five instances, showing the strength of combining heuristic and exact neighborhoods.

The proposed preprocessing phase to handle obstacles, and the resulting LNS heuristic can easily be adapted to a number of more general network design problems. A first step would be to generalize it to handle more topologies, for example the looped topology.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in “Synthetic instances for large off-shore wind farm layout” at <http://dx.doi.org/10.11583/DTU.13134731>. This set of instances has been created from a mix of public data and data generated based on realistic assumptions. The data are described in: D. Cazzaro and D. Pisinger, “Variable neighborhood search for large off-shore wind farm layout optimization.”

REFERENCES

- [1] S. M. Alagab, S. Tennakoon, and C. Gould, *Review of wind farm power collection schemes*, Proc. 2015 50th Int. Univ. Power Eng. Conf. (UPEC), 2015, pp. 1–5, IEEE.
- [2] J. Bauer and J. Lysgaard, *The offshore wind farm array cable layout problem: A planar open vehicle routing problem*, J. Oper. Res. Soc. **66** (2015), no. 3, 360–368.
- [3] BVG Associates, on behalf of The Crown Estate and the Offshore Renewable Energy Catapult, *Guide to an offshore wind farm: Updated and extended*. <https://www.thecrownestate.co.uk/media/2861/guide-to-offshore-wind-farm-2019.pdf>, 2019.
- [4] D. Cazzaro, M. Fischetti, and M. Fischetti, *Heuristic algorithms for the wind farm cable routing problem*, Appl. Energy **278** (2020), 115617.
- [5] D. Cazzaro and D. Pisinger, *Variable neighborhood search for large offshore wind farm layout optimization*, Comput. Oper. Res. **138** (2022), 105588.
- [6] G. Clarke and J. W. Wright, *Scheduling of vehicles from a central depot to a number of delivery points*, Oper. Res. **12** (1964), no. 4, 568–581.
- [7] K. Das and N. Antonios Cutululis, *Offshore wind power plant technology catalogue—Components of wind power plants, AC collection systems and HVDC systems*, Technical report, Danmarks Tekniske Universitet, October 2017.
- [8] M. Fischetti, J. R. Kristoffersen, T. Hjort, M. Monaci, and D. Pisinger, *Vattenfall optimizes offshore wind farm design*, INFORMS J. Appl. Anal. **50** (2020), no. 1, 80–94.
- [9] M. Fischetti and D. Pisinger, *Optimal wind farm cable routing: Modeling branches and offshore transformer modules*, Networks **72** (2018), no. 1, 42–59.
- [10] M. Fischetti and D. Pisinger, *Optimizing wind farm cable routing considering power losses*, Eur. J. Oper. Res. **270** (2018), no. 3, 917–930.
- [11] X. Gong, S. Kuenzel, and B. C. Pal, *Optimal wind farm cabling*, IEEE Trans. Sustain. Energy **9** (2017), no. 3, 1126–1136.
- [12] A. G. Gonzalez-Rodriguez, *Review of offshore wind farm cost components*, Energy Sustain. Develop. **37** (2017), 10–19.
- [13] P. Hou, J. Zhu, K. Ma, G. Yang, W. Hu, and Z. Chen, *A review of offshore wind farm layout optimization and electrical system design methods*, J. Modern Power Syst. Clean Energy **7** (2019), no. 5, 975–986.
- [14] A. Jenkins, M. Scutariu, and K. Smith, *Offshore wind farm inter-array cable layout*, Proc. 2013 IEEE Grenoble Conf., 2013, pp. 1–6, IEEE.
- [15] R. Jin, P. Hou, G. Yang, Y. Qi, C. Chen, and Z. Chen, *Cable routing optimization for offshore wind power plants via wind scenarios considering power loss cost model*, Appl. Energy **254** (2019), 113719.
- [16] N. Katayama, *A combined fast greedy heuristic for the capacitated multicommodity network design problem*, J. Oper. Res. Soc. **70** (2019), no. 11, 1983–1996.
- [17] G. Katsouris, *Infield cable topology optimization of offshore wind farms*, Master's thesis, Delft University of Technology, Netherlands, 2015.
- [18] S. Lundberg, *Wind farm configuration and energy efficiency studies-series DC versus AC layouts*, Department of Energy and Environment; Chalmers University of Technology; SE-412 96 Göteborg; Sweden, 2006.
- [19] J.-A. Pérez-Rúa and N. A. Cutululis, *Electrical cable optimization in offshore wind farms—a review*, IEEE Access **7** (2019), 85796–85811.
- [20] A. Pillai, J. Chick, L. Johanning, M. Khorasanchi, and V. De Laleu, *Offshore wind farm electrical cable layout optimization*, Eng. Optim. **47** (2015), no. 12, 1689–1708.
- [21] Y. Qi, P. Hou, G. Liu, R. Jin, Z. Yang, G. Yang, and Z. Dong, *Cable connection optimization for heterogeneous offshore wind farms via a Voronoi diagram based adaptive particle swarm optimization with local search*, Energies **14** (2021), no. 3, 644.
- [22] L. Ramírez, D. Fraile, and G. Brindley, *Offshore wind in Europe: Key trends and statistics 2019*, Technical report, WindEurope, 2020. <https://windeurope.org/wp-content/uploads/files/about-wind/statistics/WindEurope-Annual-Offshore-Statistics-2019.pdf>
- [23] R. A. Walling and T. Ruddy, *Economic optimization of offshore wind farm substations and collection systems*, Proc. 5th Int Worksh. Large Scale Integr Wind Power Transm. Netw. Offshore Windfarms, 2005, pp. 396–402.
- [24] X. Yi, M. Scutariu, and K. Smith, *Optimisation of offshore wind farm inter-array collection system*, IET Renew. Power Gener. **13** (2019), no. 11, 1990–1999.

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APPENDIX A: COMPARISON OF BALANCED AND UNBALANCED CONFIGURATIONS

A.1 | BRANCHED TOPOLOGY

TABLE A1 Comparison of the final solution costs for balanced and unbalanced configurations for the branched topology on the 15 MW synthetic instances

Instance	Balanced (M€)	Unbalanced (M€)	Delta (M€)	Percentage (%)
A	20.31	18.88	−1.4	−7.0
B	48.55	48.00	−0.5	−1.1
C	46.82	46.25	−0.6	−1.2
D	84.40	83.86	−0.5	−0.6
E	65.03	64.91	−0.1	−0.2
F	83.34	82.66	−0.7	−0.8
G	66.19	66.19	0.0	0.0
H	98.46	98.19	−0.3	−0.3
I	175.44	175.35	−0.1	−0.1
J	113.16	112.02	−1.1	−1.0
Average			−0.5	−1.2

TABLE A2 Comparison of the final solution costs for balanced and unbalanced configurations for the branched topology on the 10 MW synthetic instances

Instance	Balanced (M€)	Unbalanced (M€)	Delta (M€)	Percentage (%)
A	21.62	20.79	−0.8	−3.8
B	54.74	54.08	−0.7	−1.2
C	53.49	51.83	−1.7	−3.1
D	97.14	96.07	−1.1	−1.1
E	74.36	71.98	−2.4	−3.2
F	95.43	94.09	−1.3	−1.4
G	78.15	75.41	−2.7	−3.5
H	117.41	114.10	−3.3	−2.8
I	205.90	202.51	−3.4	−1.6
J	130.02	127.20	−2.8	−2.2
Average			−2.0	−2.4

A.2 | RADIAL TOPOLOGY

TABLE A3 Comparison of the final solution costs for balanced and unbalanced configurations for the radial topology on the 15 MW synthetic instances

Instance	Balanced (M€)	Unbalanced (M€)	Delta (M€)	Percentage (%)
A	20.56	19.82	−0.7	−3.6
B	48.98	48.50	−0.5	−1.0
C	47.43	47.02	−0.4	−0.9
D	85.98	84.97	−1.0	−1.2
E	67.05	67.13	0.1	0.1
F	85.52	84.39	−1.1	−1.3
G	66.56	66.56	0.0	0.0
H	99.38	99.38	0.0	0.0
I	175.98	175.89	−0.1	−0.1
J	114.81	113.67	−1.1	−1.0
Average			−0.5	−0.9

TABLE A4 Comparison of the final solution costs for balanced and unbalanced configurations for the radial topology on the 10 MW synthetic instances

Instance	Balanced (M€)	Unbalanced (M€)	Delta (M€)	Percentage (%)
A	22.49	21.71	−0.8	−3.5
B	55.00	54.96	0.0	−0.1
C	54.76	52.91	−1.9	−3.4
D	98.37	98.15	−0.2	−0.2
E	75.57	73.87	−1.7	−2.2
F	96.51	95.45	−1.1	−1.1
G	78.82	76.73	−2.1	−2.7
H	118.20	116.68	−1.5	−1.3
I	207.39	204.18	−3.2	−1.5
J	131.52	130.76	−0.8	−0.6
Average			−1.3	−1.7